

# Richin Mrudul

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## EDUCATION

### Purdue University

West Lafayette, IN

**B.S. in Computer Science; Minor in Math; Concentration in Machine Intelligence**

Expected December 2027

Relevant Courses: Data Structures & Algorithms, Systems Programming, Computer Architecture, Data Mining and Machine Learning, Discrete Math, Statistics, Linear Algebra, and Multivariate Calculus

## TECHNICAL SKILLS

**Languages:** Go, C/C++/C#, Java, Python, TypeScript, JavaScript, SQL, Bash

**Libraries/Frameworks:** React, React Native, Node.js, Next.js, FastAPI, JUnit, PyTorch, TensorFlow, scikit-learn

**Technologies & Tools:** REST API, Git, Docker, Kubernetes, Azure AKS, AWS, GCP, CI/CD, GitHub Actions, GHCR, PostgreSQL, MySQL, Redis, MinIO/S3, FFmpeg, Prometheus, Grafana, Django, Netlify, Render

## EXPERIENCE

### Pendo.io

Raleigh, North Carolina

*Software Engineer Intern*

June 2026 - Present

- Summer 2026

### Boilerexams

West Lafayette, IN

*Backend Developer*

August 2025 - Present

- Develop backend systems for Boilerexams, serving **30,000+ students across 11M+ exam submissions**
- Optimized PostgreSQL queries across 11M+ submissions and 3,000+ questions, reducing latency by **20%**
- Migrated API endpoints from **TypeScript/Prisma to Go** for faster, more maintainable services
- Containerized backend services with **Docker and structured logging and error handling**

### Profitize

Remote

*Software Engineer*

July 2025 - Present

- Built a SaaS analytics platform for **500+ Amazon sellers** automating profit analysis and product research
- Engineered scraping pipeline with **Python, Playwright, Selenium, & RapidFuzz**, reducing research time **80%**
- Implemented real-time fuzzy matching across Amazon listings to improve product data accuracy
- Deployed infrastructure using **Netlify, Render, GitHub Actions**, achieving **under 2-second page-load times**

### The Data Mine - Purdue University

West Lafayette, IN

*Undergraduate Data Science Researcher*

August 2024 - May 2025

- Built an **AI-assisted data pipeline** for the **Indiana Soybean Alliance**, cutting collection time **96%** (428→14s)
- Implemented **LLM-assisted program classification** to reduce manual categorization of collected records
- Migrated hard-coded datasets to a live **PostgreSQL** database enabling real-time updates and search
- Developed an admin dashboard for publishing and managing program data through a scalable backend

## PROJECTS

### Distributed Video Processing Infrastructure [🔗](#) | [Demo](#) [🔗](#)

- Built **Kubernetes**-deployed video-processing system with **FastAPI, Redis/RQ, Postgres**, and **MinIO**
- Scaled workers **1→5** in benchmarks, increasing throughput **2.5x** from **4.46** to **11.13 uploads/sec**
- Separated upload ingestion from CPU-bound **FFmpeg** execution via async queues and worker pools
- Improved reliability with **idempotency, retries**, stuck-job recovery, and admission control

### AgentGuard [🔗](#)

- Built and released AgentGuard **v0.2.0**, an **open-source CLI** for repeatable coding-agent evaluations
- Piloted AgentGuard with a collaborator; observed **40% fewer unsafe issues** in comparable local runs
- Detected **8 unsafe-behavior classes** with runtime guardrails, post-run validation, traces, and replay
- Shipped **990 passing tests**, CI gates, static dashboards, smoke validation, and 4 report/export formats

### FitTrack [🔗](#) | [Website](#) [🔗](#)

- Built a full-stack fitness tracker using **React** and **Firebase** for workout logging and progress tracking
- Implemented onboarding with **Firebase Auth** and **Google OAuth** to personalize plans from user fitness data
- Integrated **OpenAI GPT-4** to generate personalized **7-day workout plans**, improving completion **45%**
- Used by **20+ users** generating **100+ plans** and logging **500+ workouts** in the first **3 weeks**